2.2 Inductive and Deductive Reasoning

Name _____ Period ____ Date _____

Geometry 2.2 (30-minute plan) · Student sheet

I can…

- use inductive reasoning to describe a pattern and make a conjecture.
- show a conjecture is false with a counterexample.
- use the Laws of Detachment and Syllogism to reach a conclusion.

Vocabulary · fill in the missing word

Conjecture — an unproven statement that is based on _____

Inductive reasoning — finding a pattern in specific cases and then writing a _____ for the general case

Counterexample — a specific case for which a _____ is false

Deductive reasoning — using facts, _____, accepted properties, and the laws of logic to form a logical argument

Part 1 · Notes and examples (fill in each note, then work the example)

Inductive reasoning: look at examples, find a _____, and make a conjecture.

Example 1

Describe the pattern and find the next number: 3, 7, 11, 15, …

Counterexample: to show a conjecture is false, you need only _____ counterexample.

Example 2

Find a counterexample: “The difference of two positive numbers is always positive.”

Law of Detachment: if $p \rightarrow q$ is true and p is true, then _____ is true.

Law of Syllogism: if $p \rightarrow q$ and $q \rightarrow r$ are true, then _____ is true.

Example 3

What can you conclude? “If two segments have the same length, then they are congruent. AB = CD = 5.”

Part 2 · Partner practice

Partner practice

Write the statement that follows: “If a figure is a square, then it is a rectangle.” “If a figure is a rectangle, then it has four right angles.”